Dynamic Compatibility and Reorganization Across Nonlinear Systems

Abstract

This study investigates whether structurally different nonlinear systems may exhibit comparable compatibility-dependent reorganization signatures across chaotic, oscillatory, adaptive, collective, and distributed lattice regimes. The analyzed systems include Lorenz, Rossler, Duffing, Van der Pol, Kuramoto, Stuart-Landau, and coupled map lattice dynamics. Comparative analyses were supported by bootstrap resampling, shuffle randomization, surrogate or null controls, endpoint-aware robustness procedures, normalized overlays, and cross-system persistence metrics. Observed signatures included persistence enhancement, reduced transition activity, local stabilization, synchronization restructuring, metastability, adaptive recovery, and distributed collective organization. The framework is interpreted as operational and comparative rather than universal or ontological.

1. Introduction

Nonlinear systems are often studied within isolated mathematical frameworks. This is effective for system-specific understanding, but it can obscure organizational signatures that may recur across different dynamical regimes.

The present work asks a narrower comparative question: can structurally different systems exhibit comparable compatibility-dependent reorganization signatures when analyzed using a shared operational framework?

The goal is not to claim equivalence between systems or to propose a universal law. Instead, we examine whether persistence enhancement, local stabilization, reorganization windows, synchronization restructuring, metastability, and distributed organization appear across chaotic, oscillatory, adaptive, collective, and lattice-based systems.

Because cross-system comparisons are vulnerable to normalization artifacts and geometry-only explanations, robustness procedures are treated as part of the central analysis rather than as a secondary appendix.

This work extends a previous minimal agent-based study of compatibility-dependent reorganization (Wygrabek, 2026, https://doi.org/10.5281/zenodo.19563941) by examining whether comparable organization signatures emerge across structurally different nonlinear systems.

2. Methods and workflow

2.1 Comparative systems

The review-candidate version integrates evidence from classical chaotic systems, oscillator systems, collective synchronization models, adaptive oscillator networks, and coupled map lattice dynamics.

System class	Systems	Role in the paper
Chaotic attractors	Lorenz, Rossler, Duffing	Baseline compatibility-dependent restructuring and dwell/persistence effects
Oscillatory dynamics	Van der Pol	Bootstrap stability and local/global analysis in continuous oscillator dynamics
Collective synchronization	Kuramoto	Compatibility-dependent synchronization transition and

		topology robustness
Adaptive collective oscillators	Stuart-Landau	Metastability, hysteresis, recovery dynamics, and adaptive restructuring
Distributed lattice dynamics	CLM	Distributed cluster organization, epsilon-star windows, and collective map dynamics

2.2 Metrics and controls

Operational compatibility definition: Compatibility is treated operationally rather than ontologically. Compatibility-dependent regions refer to parameter intervals in which local interactions produce increased persistence, reduced transition activity, synchronization restructuring, metastability, or localized stabilization relative to neighboring parameter regimes and controls.

Operational reorganization score clarification: The reorganization score is treated as an operational composite indicator intended to capture changes in persistence, restructuring activity, synchronization behavior, metastability, or cluster formation relative to neighboring parameter regions and null controls.

The analysis uses operational compatibility thresholds, persistence metrics, local/global asymmetry, transition activity, synchronization measures, reorganization scores, metastability indicators, and recovery-related metrics.

Validation procedures include shuffle randomization, null or surrogate controls, bootstrap resampling, endpoint-aware checks, and topology or coupling-structure robustness analyses.

Computational workflows, plotting, and document integration included AI-assisted support under continuous human supervision and validation.

The central comparison is not numerical identity between systems. It is the recurrence of related organizational signatures across different regimes under compatibility-dependent constraints.

2.3 Human-AI collaborative research workflow

Portions of the computational workflow, including code drafting, figure integration, and document organization, were AI-assisted under continuous human supervision and validation.

The study was conducted using an iterative human-AI collaborative workflow. Research direction, comparative reasoning, interpretation, validation priorities, and synthesis decisions remained human-guided. AI-assisted systems supported code generation, robustness organization, figure integration, document consolidation, and consistency checking. The workflow was iterative rather than single-prompt based, and candidate results were repeatedly evaluated through controls, recovery procedures, cross-system comparisons, and document-level refinement.

3. Main figure architecture

Figure	**Role**	Status
Figure 1	Unified synthesis schematic	Integrated
Figure 2	CLM reorganization window	Integrated
Figure 3	Local/global asymmetry	Integrated
Figure 4	Shuffle/null control comparison	Integrated
Figure 5	Bootstrap stability	Integrated
Figure 6	Kuramoto reorganization/topology	Integrated

	robustness	
Figure 7	Kuramoto synchronization transition	Integrated
Figure 8	Stuart-Landau hysteresis	Integrated
Figure 9	Stuart-Landau metastability landscape	Integrated
Figure 10	Stuart-Landau recovery dynamics	Integrated

4. Results

4.1 Unified synthesis of the comparative framework

The synthesis schematic summarizes the relationship between the analyzed system classes, observed signatures, validation procedures, and interpretation boundaries. It is included as a visual map for the review-candidate version.

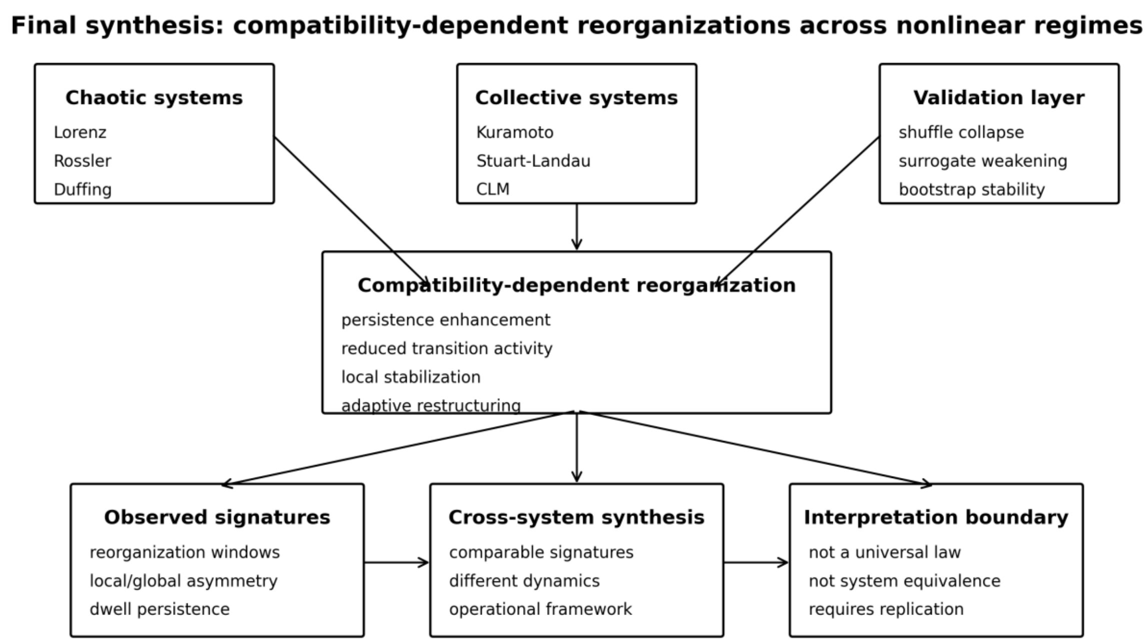

Figure 1. Schematic synthesis of the comparative framework. The framework connects chaotic systems, collective systems, validation procedures, and compatibility-dependent reorganization signatures. The interpretation remains operational and comparative rather than universal or ontological.

4.2 Distributed lattice reorganization in CLM dynamics

Coupled map lattice analyses provide a distributed, discrete collective regime. Unlike low-dimensional trajectory systems, CLM extends the framework toward spatially distributed collective restructuring dynamics. Small epsilon values produce fragmented or weakly coupled organization, while excessively large epsilon values promote rigid synchronization. Intermediate regions near epsilon ≈ 0.18 displayed the strongest distributed reorganization signatures across bootstrap-supported analyses.

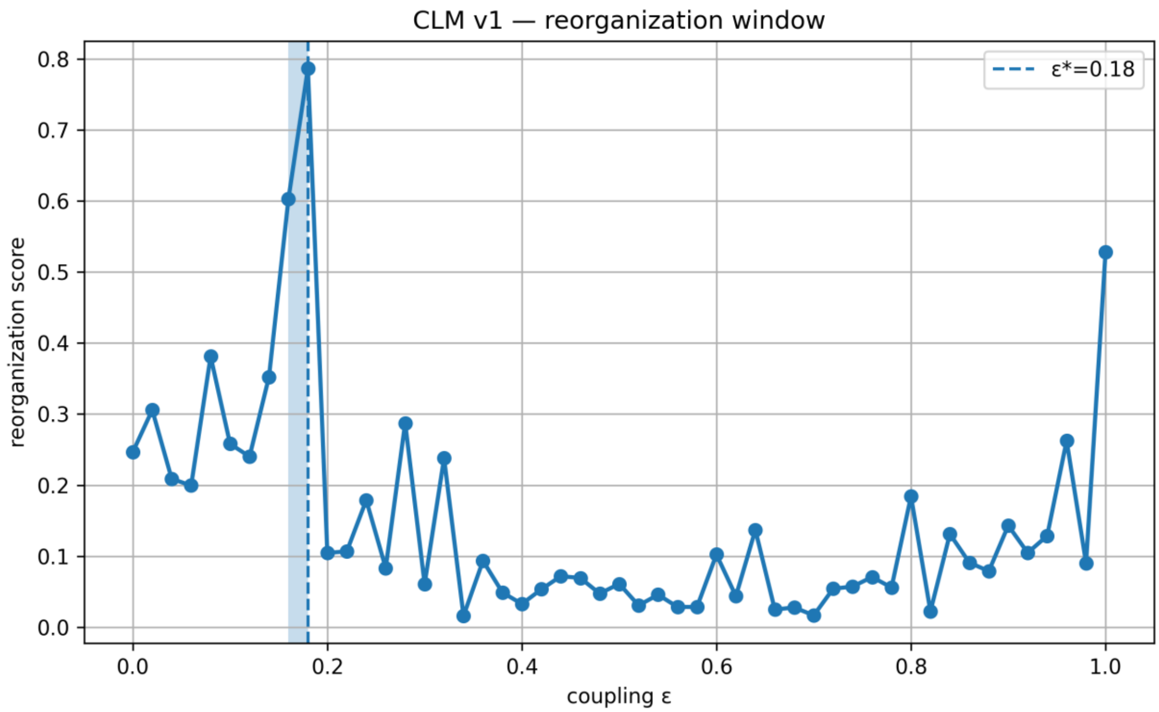

Figure 2. CLM reorganization window. The figure summarizes distributed collective reorganization under changing coupling/epsilon conditions and supports the extension of the framework beyond continuous chaotic or oscillator systems.

4.3 Local/global asymmetry and local stabilization

Local/global asymmetry provides a mechanism-oriented diagnostic. When local organization exceeds expectations from global geometry alone, the result supports the interpretation of localized stabilization rather than purely static phase-space structure.

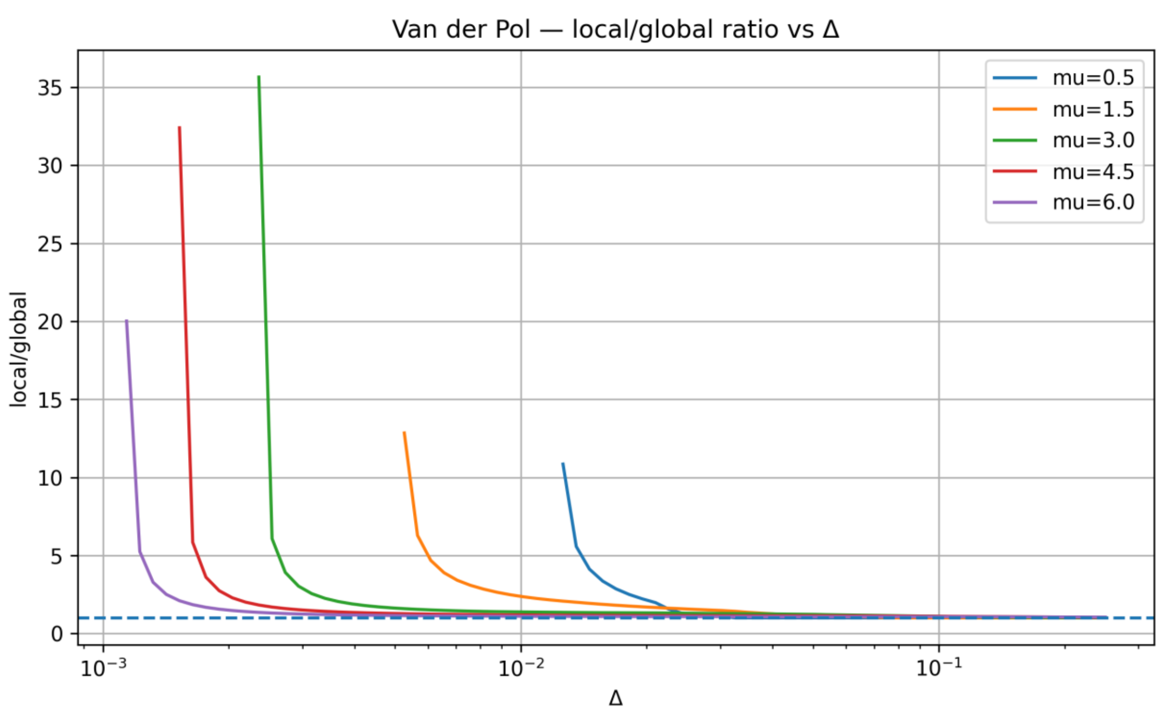

Figure 3. Local/global asymmetry. Local organization exceeds global geometric expectations in selected oscillator analyses, supporting the role of localized stabilization under compatibility-dependent constraints.

4.4 Shuffle and null control behavior

Shuffle and null procedures weakened persistence-related organization selectively rather than uniformly across the parameter range.

Shuffle and null controls test whether observed persistence and reorganization signatures survive disruption of relational or temporal structure. Suppression of structure under randomization strengthens the argument that the effect is not purely geometric.

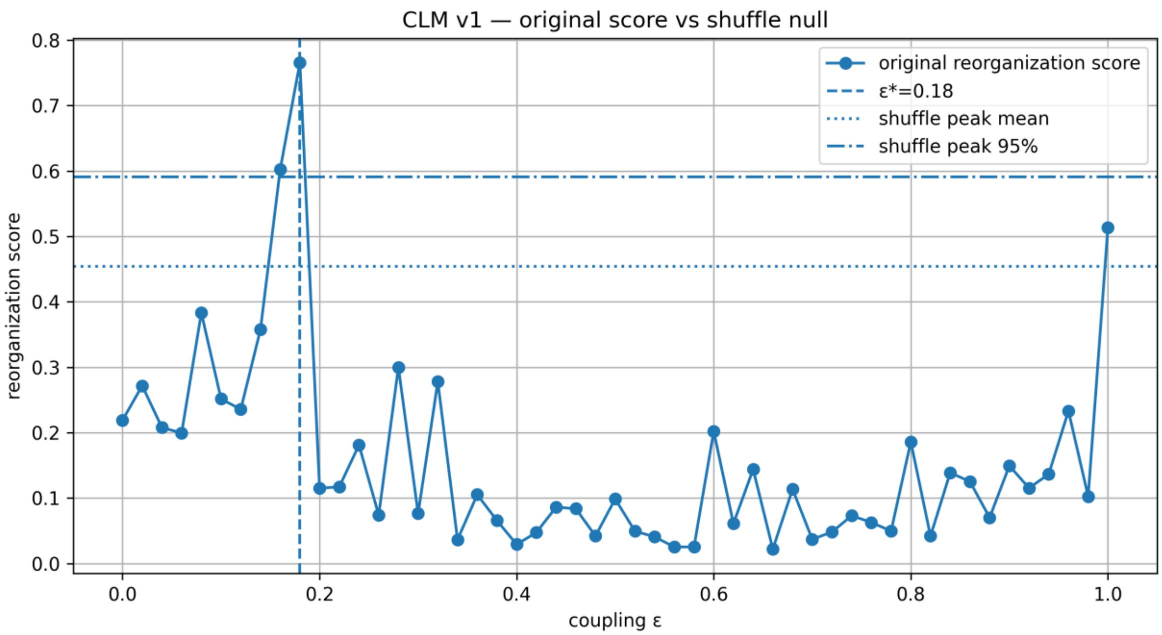

Figure 4. Original versus shuffle/null comparison. Randomization weakens persistence and reorganization signatures, indicating sensitivity to relational organization rather than only static distributional geometry.

4.5 Bootstrap stability of characteristic windows

Bootstrap analyses (N = 1000 resampled realizations) indicated that the characteristic epsilon-star region remained localized near epsilon ≈ 0.18.

Bootstrap analyses test whether characteristic windows remain detectable under repeated resampling. Stability under resampling does not prove universality, but it strengthens the operational robustness of the observed window structure.

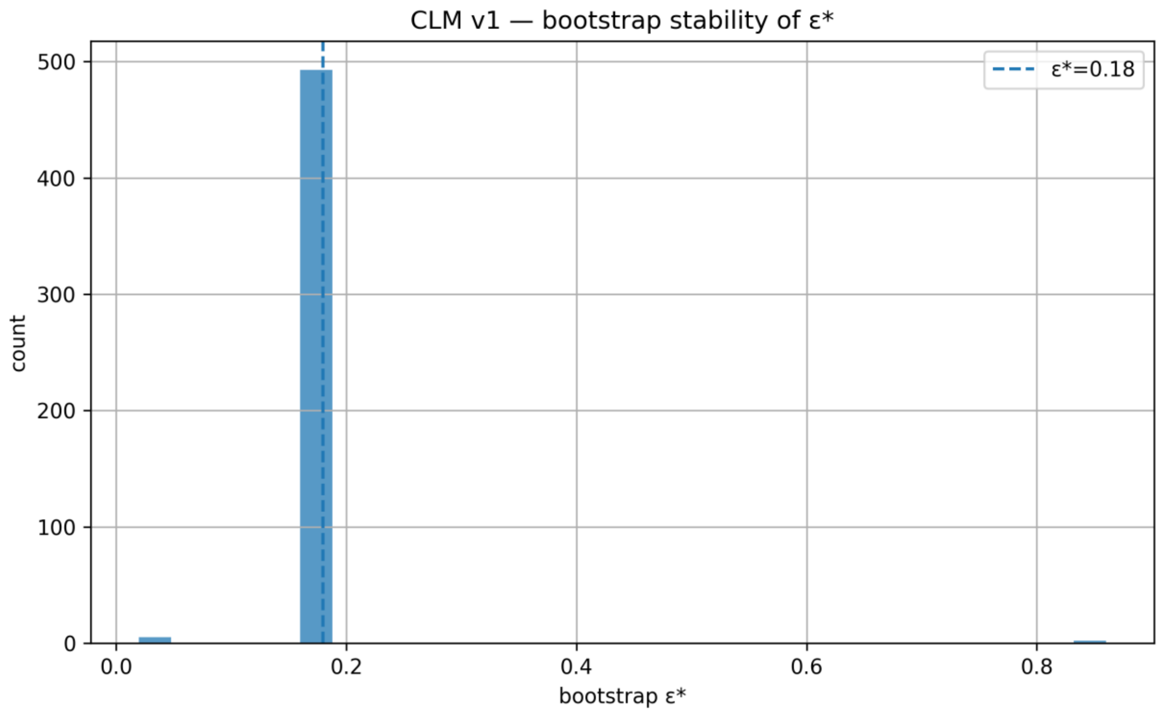

Figure 5. Bootstrap stability of the characteristic epsilon-star window. Repeated resampling supports the detectability of the characteristic CLM reorganization window while preserving the need for normalization caution.

4.6 Kuramoto topology robustness and reorganization score

Kuramoto analyses provide the transition from individual or trajectory-level dynamics to collective synchronization. Topology robustness tests assess whether reorganization signatures depend on a single network configuration.

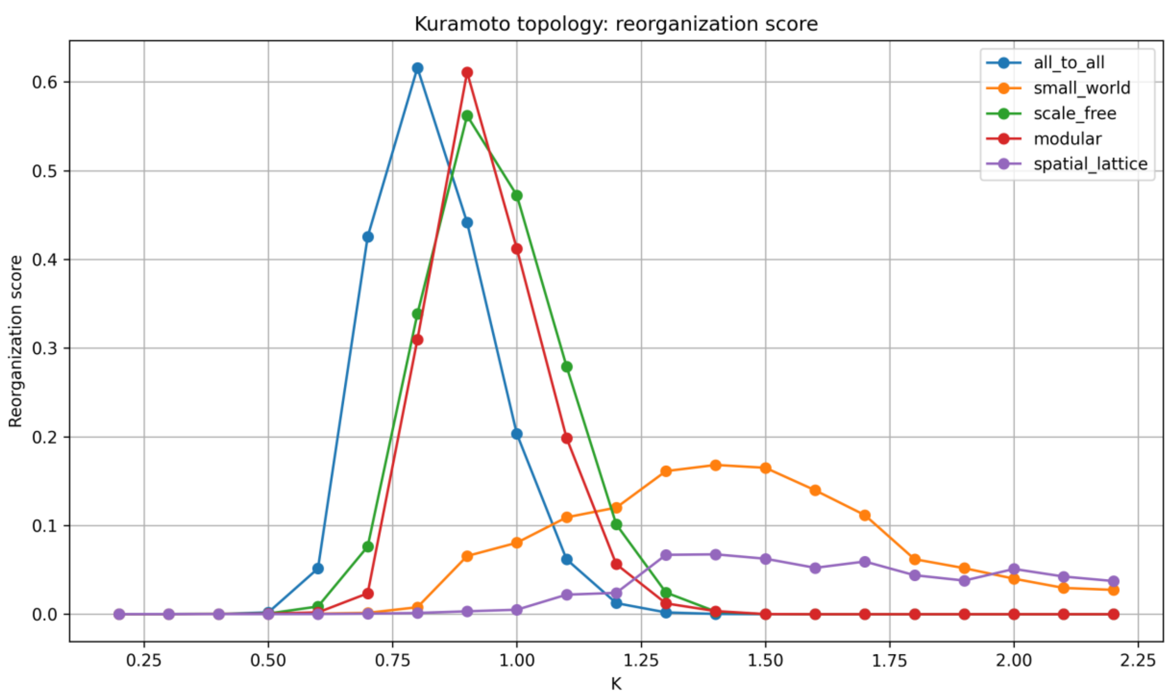

Figure 6. Kuramoto topology robustness and reorganization score. Compatibility-dependent reorganization signatures remain visible across topology-related collective synchronization tests.

4.7 Kuramoto synchronization transition

Synchronization measures provide a direct collective-order diagnostic. The Kuramoto layer supports the extension of the framework from chaotic dynamics to collective coherence formation.

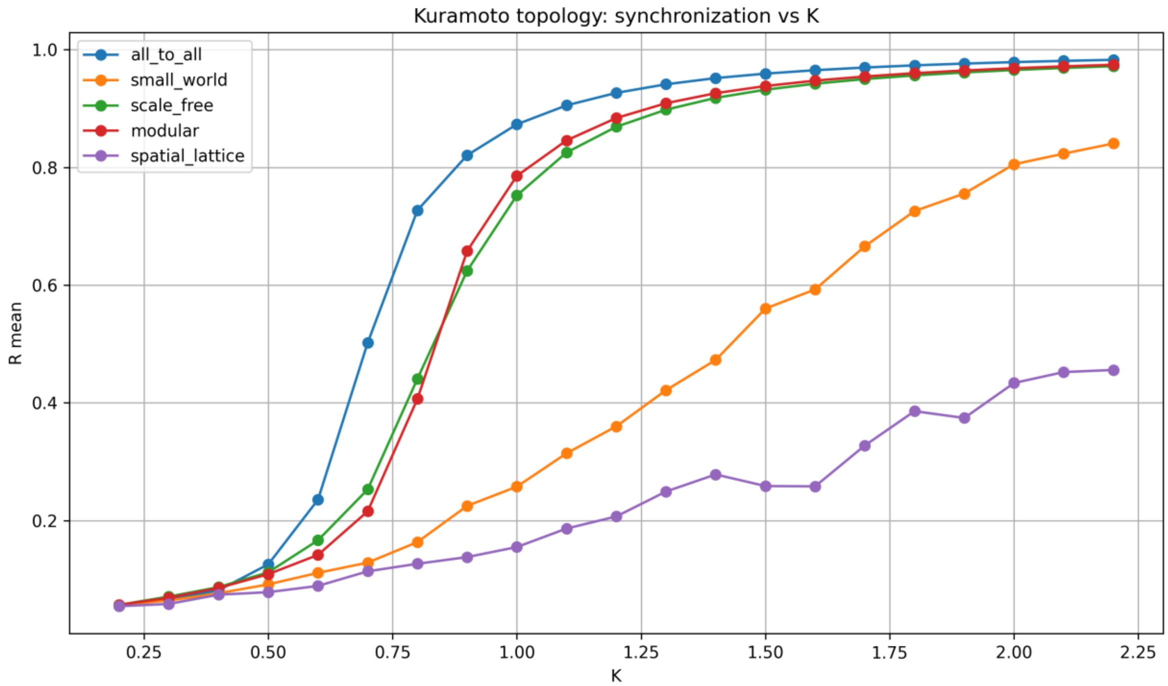

Figure 7. Kuramoto synchronization transition. The R-versus-K structure shows collective coherence formation across coupling conditions and provides a synchronization-level counterpart to compatibility-dependent reorganization windows.

4.8 Stuart-Landau hysteresis and adaptive restructuring

Stuart-Landau systems add adaptive collective oscillator dynamics, including hysteresis, recovery effects, and memory-like behavior. These results broaden the framework beyond static synchronization transitions.

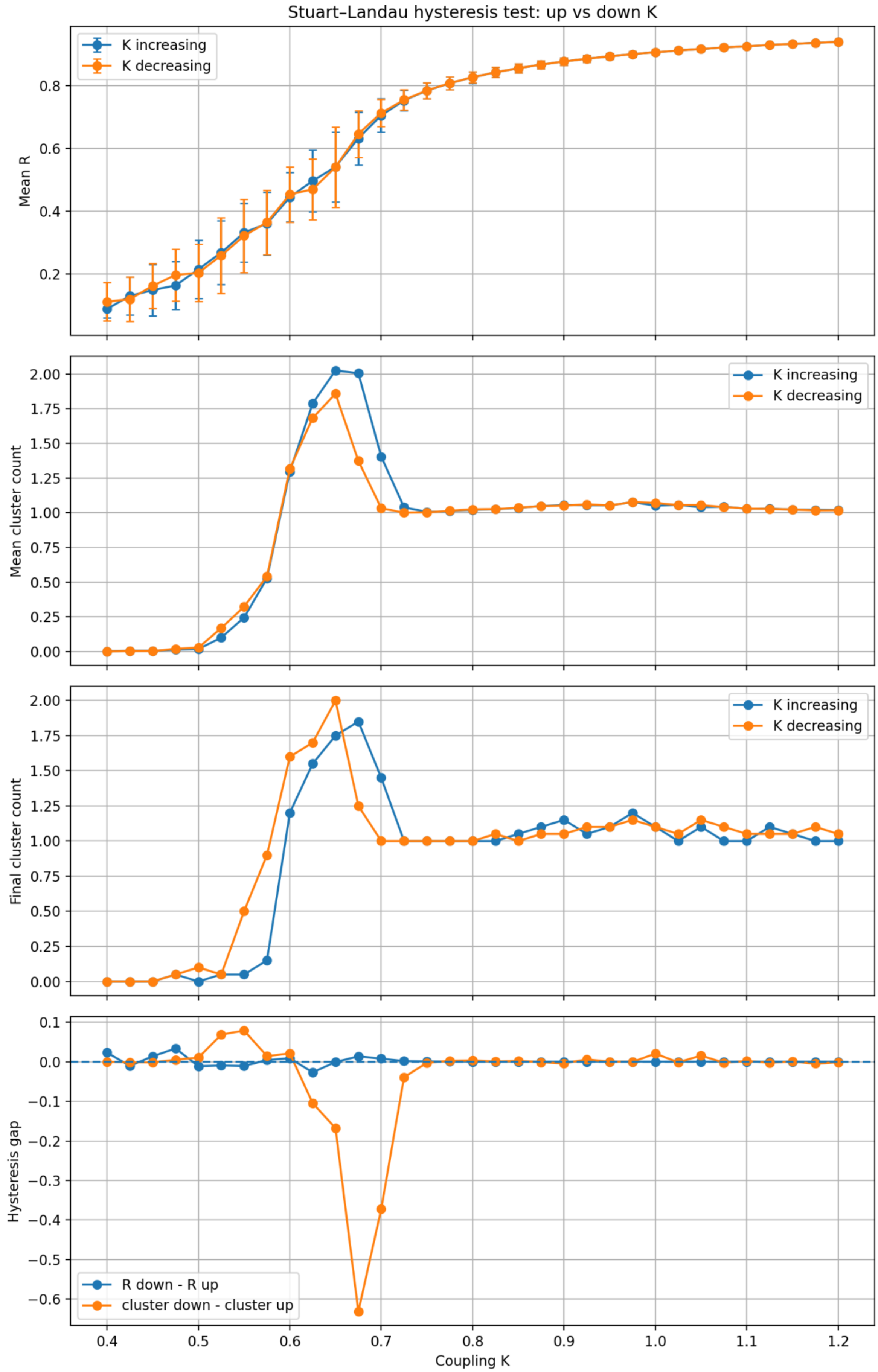
Stuart–Landau hysteresis test: up vs down K
K increasing
K decreasing
Mean R
Mean cluster count
Final cluster count
Hysteresis gap
R down - R up
cluster down - cluster up
Coupling K
0.4
0.5
0.6
0.7
0.8
0.9
1.0
1.1
1.2

Figure 8. Stuart-Landau hysteresis test. Hysteresis and adaptive restructuring indicate that compatibility-dependent transitions may depend on history, recovery pathway, and collective state memory.

4.9 Stuart-Landau metastability landscape

Metastability is a central collective-system signature. The landscape view supports the interpretation of intermediate organizational regimes rather than only binary synchronized/desynchronized states.

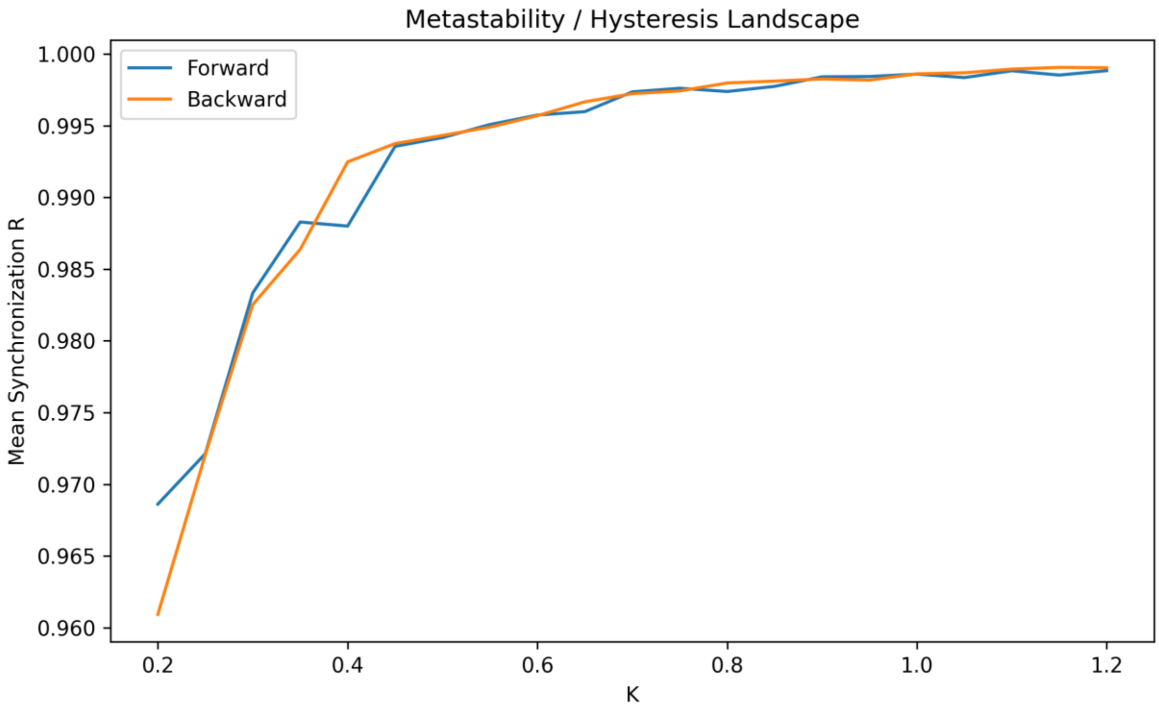

Figure 9. Stuart-Landau metastability landscape. Metastable regions provide a collective adaptive counterpart to persistence-enhancement and local stabilization observed in other systems.

4.10 Recovery dynamics

Recovery dynamics test whether a system returns to its prior organizational state after perturbation or transition. This adds a temporal resilience dimension to the comparative framework.

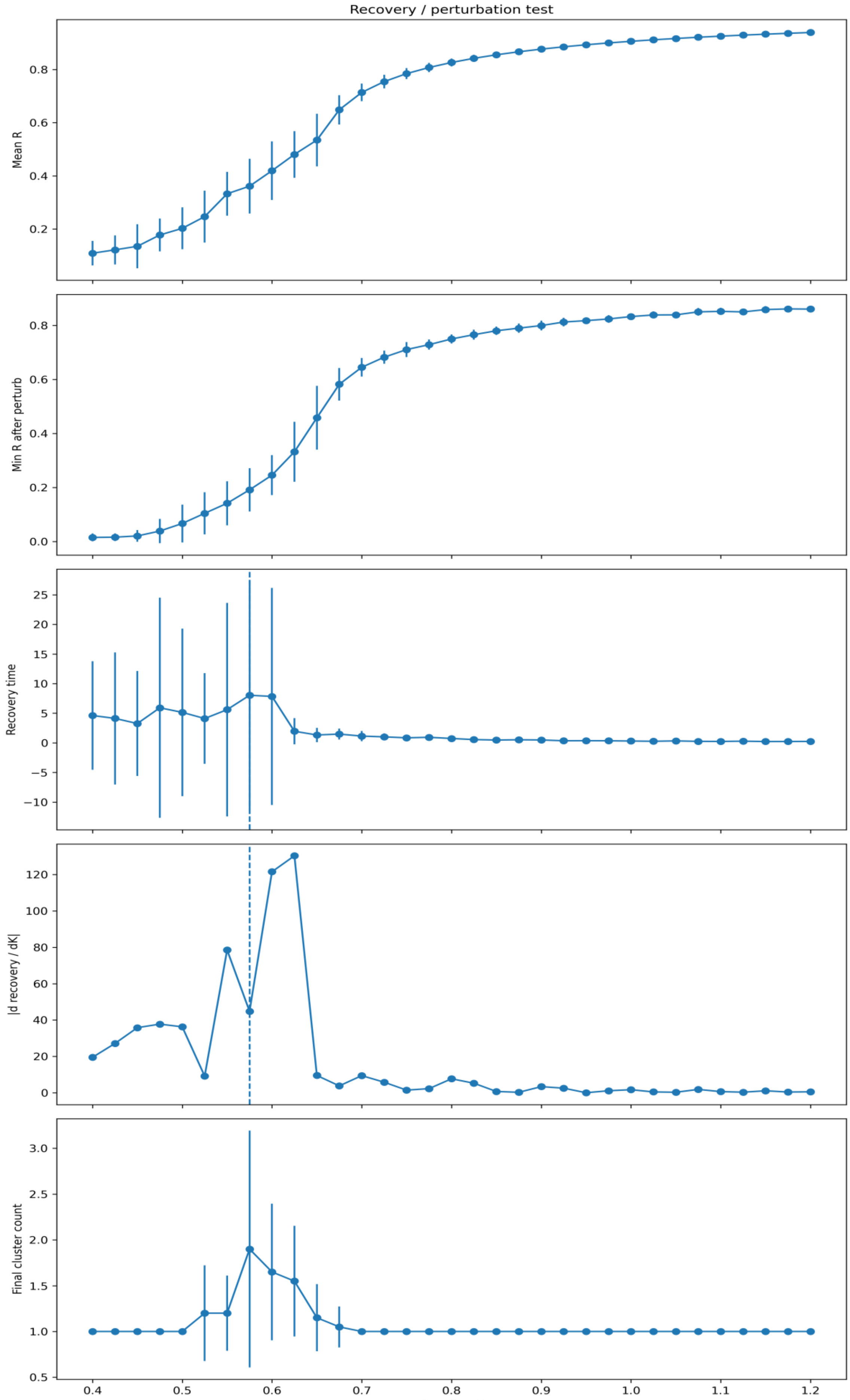
Recovery / perturbation test
Mean R
Min R after perturb
Recovery time
|d recovery / dK|
Final cluster count
Coupling K
0.4
0.5
0.6
0.7
0.8
0.9
1.0
1.1
1.2

Figure 10. Stuart-Landau recovery dynamics. Recovery tests show delayed or structured return behavior near transition regions, supporting adaptive recovery dynamics in collective systems. The coupling K axis is shared across all subpanels. Relative recovery values may appear shifted because the metric is represented as a relative recovery-scale displacement rather than absolute physical time.

5. Minimum quantitative summary tables

The current review-candidate version includes two minimum quantitative tables. These tables should be treated as summary-level orientation tables; full numerical scans remain in the supplementary archive.

System	**Primary signature**	**Robustness/validation role**	**Interpretation boundary**
Lorenz	Transition-driven chaotic restructuring	Shuffle/surrogate comparison	Baseline chaotic benchmark
Rossler	Dwell and transition restructuring	Shuffle/surrogate comparison	Secondary chaotic benchmark
Duffing	Strong bistable persistence	Shuffle and dwell contrast	Strongest bistability-oriented case
Van der Pol	Stable delta-star behavior	Bootstrap and local/global analysis	Oscillatory validation layer
Kuramoto	Synchronization restructuring	Topology and coupling robustness	Collective synchronization layer
Stuart-Landau	Metastability, hysteresis, recovery	Null/topology/recovery tests	Adaptive collective dynamics layer
CLM	Distributed cluster reorganization	Bootstrap and shuffle/null controls	Distributed lattice extension
Potential reviewer objection		Reviewer response layer	
This may be normalization-dependent		Normalization sensitivity is explicitly treated as a limitation; full scans remain supplementary	
This may be geometry-only		Shuffle, surrogate/null, and dwell/persistence controls address relational and temporal structure	
Systems are too different		The paper does not claim equivalence; it compares operational signatures	
The framework is not formal enough		The paper is positioned as exploratory comparative dynamics, not a final mathematical theory	
AI involvement may reduce trust		Human-AI workflow is disclosed as human-guided and methodologically transparent	

6. Robustness and interpretation

Shuffle randomization consistently reduced persistence structure and weakened local stabilization signatures across multiple system classes.

Surrogate or null procedures partially preserved geometry or distributional structure while weakening coherent persistence enhancement and adaptive restructuring signatures.

Bootstrap analyses indicated that characteristic reorganization windows remained detectable across repeated resampling procedures in selected systems, although sensitivity to normalization and parameterization remains present.

Endpoint exclusion analyses and topology robustness tests reduce, but do not eliminate, concerns about finite-window artifacts and topology dependence.

7. Discussion

Mechanistic sketch: Compatibility-dependent constraints may reduce dynamically accessible pathways, increasing the probability of temporary persistence, localized stabilization, synchronization restructuring, or metastable organization across nonlinear regimes.

The results support an operational comparative interpretation: structurally different nonlinear systems may display comparable compatibility-dependent reorganization signatures without being mathematically or physically equivalent.

The CLM regime extends the framework from low-dimensional trajectory restructuring toward distributed collective organization. This suggests that compatibility-dependent reorganization signatures may remain detectable not only in individual trajectories or synchronization dynamics, but also in lattice-like spatial collective systems.

The strongest contribution of the present work is not a single curve or a single system-specific result. It is the cross-system organization of evidence across chaotic, oscillatory, adaptive, collective, and distributed lattice regimes.

The framework remains exploratory. It should be interpreted as comparative evidence for recurring reorganization signatures, not as a universal law or a completed theory of nonlinear organization.

8. Limitations

Normalization clarification: Cross-system comparisons used operational normalization procedures intended to preserve relative organizational structure rather than absolute equivalence.

Cross-system comparisons remain sensitive to normalization strategy, operational definitions, parameter selection, and metric design.

Topology dependence remains an open issue for collective systems. Additional network classes and parameter scans are required.

Surrogate-memory ambiguity remains possible: some controls may preserve partial temporal structure while suppressing other forms of organization.

Independent replication and external review are required before broader generality claims can be made.

9. Supplementary roadmap

Supplement	Contents
Supplement A	Extended chaos overlays for Lorenz, Rossler, and Duffing
Supplement B	Bootstrap, shuffle, surrogate/null, and endpoint-exclusion robustness scans
Supplement C	Kuramoto and Stuart-Landau collective-system scans, topology tests, recovery dynamics, and metastability maps
Supplement D	CLM distributed lattice scans, cluster persistence, epsilon-star bootstrap, and exploratory perturbation groundwork

10. Reference backbone

1. Lorenz, E. N. (1963). Deterministic nonperiodic flow. Journal of the Atmospheric Sciences, 20(2), 130-141.
2. Rossler, O. E. (1976). An equation for continuous chaos. Physics Letters A, 57(5), 397-398.
3. Duffing, G. (1918). Erzwungene Schwingungen bei veranderlicher Eigenfrequenz und ihre technische Bedeutung. Vieweg.
4. Guckenheimer, J., & Holmes, P. (1983). Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields. Springer.
5. Strogatz, S. H. (1994). Nonlinear Dynamics and Chaos. Addison-Wesley.
6. Kuramoto, Y. (1975). Self-entrainment of a population of coupled non-linear oscillators. Lecture Notes in Physics, 39, 420-422.
7. Pikovsky, A., Rosenblum, M., & Kurths, J. (2001). Synchronization: A Universal Concept in Nonlinear Sciences. Cambridge University Press.
8. Cross, M. C., & Hohenberg, P. C. (1993). Pattern formation outside of equilibrium. Reviews of Modern Physics, 65(3), 851-1112.
9. Kaneko, K. (1992). Overview of coupled map lattices. Chaos, 2(3), 279-282.
10. Kaneko, K. (1993). Theory and Applications of Coupled Map Lattices. Wiley.
11. Efron, B. (1979). Bootstrap methods: another look at the jackknife. The Annals of Statistics, 7(1), 1-26.
12. Theiler, J., Eubank, S., Longtin, A., Galdrikian, B., & Farmer, J. D. (1992). Testing for nonlinearity in time series: the method of surrogate data. Physica D, 58(1-4), 77-94.

11. Conclusion

The present version integrates figures, operational comparative definitions, robustness clarification layers, distributed CLM restructuring logic, and reviewer-oriented methodological safeguards. The manuscript is intended as an exploratory comparative nonlinear dynamics preprint designed for external critique, replication-oriented discussion, and methodological refinement.

12. Quantitative Comparative Summary

System	Organizational signature	Characteristic region	Robustness behavior
Lorenz	transition restructuring	intermediate Δ window	preserved under shuffle/bootstrap
Kuramoto	synchronization restructuring	coupling transition region	topology-sensitive but persistent
Stuart–Landau	metastable restructuring	hysteresis/memory region	recovery-dependent
CLM	distributed cluster restructuring	$\varepsilon \approx 0.18$	stable across $N=1000$ bootstrap resamples